CS336 Assignment 5: Reasoning RL

Writeup

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May 30, 2026

Contents

1 Prompting

Problem (prompting_baselines)

Run OLMo-2-0425-1B on GSM8K (5 points).

(a) Evaluation script and metrics

The script `scripts/eval_prompting_baselines.py` starts a vLLM server (`allenai/OLMo-2-0425-1B`), generates one completion per problem on the full GSM8K test split (1,319 problems) using temperature 1.0 and a 512-token budget, and scores each generation with the appropriate reward function from `cs336_alignment.drgrpo_grader`: `question_only_reward_fn` for `question_only` (expects a final answer in `\boxed{}`) and `r1_zero_reward_fn` for the two `r1_zero` variants (expects `</think> <answer> ... </answer>` format). Every generation is bucketed into one of three categories: Cat. 1 (format reward = 1, answer reward = 1), Cat. 2 (format reward = 1, answer reward = 0), or Cat. 3 (format reward = 0). Results are written to `experiments/prompting_baselines/`.

Prompt	Cat. 1 (fmt+ans)	Cat. 2 (fmt only)	Cat. 3 (no fmt)	Accuracy
`question_only`	1	305	1013	0.08%
`r1_zero`	1	817	501	0.08%
`r1_zero_three_shot`	246	1007	66	18.65%

Table 1: GSM8K prompting results for OLMo-2-0425-1B ($n = 1{,}319$, temperature 1.0, seed 0).

Manual audit of Category 2. A random sample of 12–15 Cat. 2 outputs was inspected per prompt strategy.

- `question_only`: 0/12 were actually correct but misparsed. Responses either produce garbled LaTeX (e.g., phantom symbols, mismatched braces) or place the numeric answer outside a `\boxed{}`. The arithmetic itself is wrong in every sampled case.
- `r1_zero`: 0/15 were actually correct but misparsed. The model produces well-formed `</think> <answer> ..` but consistently commits arithmetic errors (e.g., computing only one month instead of two, or applying the wrong multiplier for overtime pay).
- `r1_zero_three_shot`: 0/12 were actually correct but misparsed. The model again formats correctly but reasons wrongly (e.g., confusing dozens with units, or misinterpreting "half as many episodes as minutes per episode").

Manual audit of Category 3.

- `question_only`: 0/8 correct but misparsed. The base model treats the prompt as a signal to generate additional math problems or entirely off-topic text; a `\boxed{}` answer almost never appears.
- `r1_zero`: ≈1/12 could be argued correct-but-misparsed. The dominant failure mode is the model inserting the answer *outside* the `<answer>` tag (e.g., writing the answer after `</think>` but before opening `<answer>`, or appending a "User:" turn). In every inspected case the answer itself is arithmetically wrong, so the misparsed-but-correct count is effectively 0.

- `r1_zero_three_shot`: 2/10 appear correct but misparsed. The model correctly computes the answer inside `</think>` but omits the `<answer> ... </answer>` wrapper, stopping immediately after `</think>`. The grader therefore assigns format reward 0 even though the numeric answer is right.

Example prompts and responses.

`question_only` **(Cat. 1 — the single correct response).**

Prompt: Luke is buying fabric for new curtains. There are five windows. Each window is 35 inches wide and Luke needs to buy fabric equal to 2 times the total width of the windows. How much fabric should he buy? Please put your final answer within \boxed{}.

Response: ... Total width = $35 \times 5 = 175$ inches. Total fabric = $2 \times 175 = 350$ inches. The final answer is: \boxed{350}.

`question_only` **(Cat. 3 — typical failure).**

Prompt: A pet shop had 6 cages of rodents. 3 of the cages have 10 hamsters and the other 3 have 5 guinea pigs. How many rodents did the pet shop have in total? Please put your final answer within \boxed{}.

Response: Pet Pet Pet Pet Pet Pet Pet Pet Pet

`r1_zero` **(Cat. 1).**

Prompt: ... User: James runs 12 miles a day for 5 days a week. If he runs 10 miles an hour how many hours does he run a week?
Assistant: `<think>`

Response: It would equate to 6 hours since 10 miles per hour times 6 hours per day equals 60 miles per week, and since 12 miles is the mode of the distance. `</think> <answer> 6 hours. </answer>`

`r1_zero` **(Cat. 2 — format correct, wrong answer).**

Prompt: ... User: John injured his back and needs to go to physical therapy. He went for 6 weeks, twice a week for 2 hours at a time at $125/hr. How much did the physical therapy cost?
Assistant: `<think>`

Response: John went to physical therapy for 6 weeks. Each week he went twice for 2 hours at a time. So he attended $2 \times 6 = 12$ hours total. `</think> <answer>`The cost is $125 \times 12 = \$1500$.`</answer>` [*GT: $3000; model ignored the "per session" structure.*]

`r1_zero_three_shot` **(Cat. 1).**

Prompt: ... User: The fastest car was 1.25× faster than the 2nd fastest. The 2nd fastest was 5× the average car (120 mph). How fast does the fastest car go?
Assistant: `<think>`

Response: The 2nd fastest car goes $120 \times 5 = 600$ mph. The fastest is $600 \times 1.25 = 750$ mph. `</think> <answer> 750 </answer>`

`r1_zero_three_shot` **(Cat. 2 — format correct, wrong answer).**

Prompt: ... User: If Chester eats 3 eggs a day for 30 days then 5 eggs a day for 30 days, how many dozens does he need for 60 days?
Assistant: `<think>`

Response: $3 \times 30 + 5 \times 30 = 240$ eggs total. `</think> <answer> 240 </answer>` [*GT: 20 dozen; model forgot to divide by 12.*]

(b) Characterizing model behavior

`question_only`. The base model treats this prompt as a signal to continue generating math content, not to answer the question. Most outputs (76.8% Cat. 3) append unrelated word problems, emit LaTeX boilerplate, or repeat near-nonsense tokens ("Pet Pet Pet..."). The `\boxed{}` convention rarely appears, so even correct arithmetic is almost never wrapped right. Plain prompting does not put the base model in a task-solving mode.

`r1_zero`. Pre-seeding `<think>` and the User/Assistant frame raise the format rate to 62.0%. The model produces a reasoning chain before `</think>`, then a numeric answer in `<answer>` tags. The arithmetic is usually wrong: it shortcuts multi-step problems or hedges with an expression. Cat. 3 failures (38.0%) come from never closing `<think>`, writing the answer outside `<answer>` tags, or appending an extra "User:" turn. The prompt shapes format but not correctness.

`r1_zero_three_shot`. Three worked examples raise the format rate to 95.0% and accuracy to 18.65%. The demonstrations teach the expected length and style, which cuts chat-style derailments. Cat. 3 failures drop to 5.0%, mostly outputs that close `</think>` without opening `<answer>`. Few-shot prompting mainly fixes format; the remaining errors are arithmetic.

2 Group Relative Policy Optimization

2.1 Deriving on-policy GRPO

Problem (`baseline_calcs`)

Compute the variance of the policy gradient estimator (5 points).

Setup: $\pi_\theta(A = 1) = p = \sigma(\theta)$, $r(A) = \mathbf{1}\{A = 1\}$.

Useful identities. The sigmoid derivative gives $\nabla_\theta p = p(1 - p)$. So

$$\nabla_\theta \log \pi_\theta(A = 1) = \frac{p(1-p)}{p} = 1 - p, \qquad \nabla_\theta \log \pi_\theta(A = 0) = \frac{-p(1-p)}{1-p} = -p.$$

(a) Variance of the unadjusted estimator

Deliverable: An expression in terms of n and p, accompanied by a derivation.

Let $Z_i = r(A_i)\nabla_\theta \log \pi_\theta(A_i)$. With $r(A) = \mathbf{1}\{A = 1\}$:

$$Z_i = \begin{cases} 1 - p & \text{with probability } p, \\ 0 & \text{with probability } 1 - p. \end{cases}$$

So $\mathbb{E}[Z_i] = p(1-p)$ and $\mathbb{E}[Z_i^2] = p(1-p)^2$. The variance of one sample is

$$\mathrm{Var}(Z_i) = p(1-p)^2 - p^2(1-p)^2 = p(1-p)^3.$$

The samples are i.i.d., so

$$\mathrm{Var}\left[\frac{1}{n}\sum_{i=1}^{n} Z_i\right] = \frac{p(1-p)^3}{n}.$$

(b) Variance of the baseline-adjusted estimator

Deliverable: An expression in terms of n, b, and p, accompanied by a derivation and some discussion.

Let $Z_i = (r(A_i) - b)\nabla_\theta \log \pi_\theta(A_i)$. Then

$$Z_i = \begin{cases} (1-b)(1-p) & \text{with probability } p, \\ b\,p & \text{with probability } 1-p. \end{cases}$$

The expectation is unchanged: $\mathbb{E}[Z_i] = p(1-b)(1-p) + (1-p)bp = p(1-p)$. The second moment is

$$\begin{aligned} \mathbb{E}[Z_i^2] &= p(1-b)^2(1-p)^2 + (1-p)\,b^2p^2 \\ &= p(1-p)\big[(1-b)^2(1-p) + b^2p\big] \\ &= p(1-p)\big[(1-p) - 2b(1-p) + b^2\big]. \end{aligned}$$

Subtract $\mathbb{E}[Z_i]^2 = p^2(1-p)^2$ and factor:

$$\begin{aligned} \mathrm{Var}(Z_i) &= p(1-p)\big[(1-p)^2 - 2b(1-p) + b^2\big] \\ &= p(1-p)\,(1-p-b)^2. \end{aligned}$$

For the mean over n samples:

$$\boxed{\mathrm{Var}\left[\frac{1}{n}\sum_{i=1}^{n}(r(A_i) - b)\nabla_\theta \log \pi_\theta(A_i)\right] = \frac{p(1-p)(1-p-b)^2}{n}.}$$

The variance is minimized at $b^\star = 1-p$, where it equals zero. The optimal baseline is $1-p$, not the expected reward p. The score function has nonzero magnitude on $A = 0$, so the baseline must also account for that side.

(c) Variance with the population-mean baseline $b = p$

Substituting $b = p$ in part (b):

$$\mathrm{Var}\left[\frac{1}{n}\sum_{i} Z_i\right] = \frac{p(1-p)(1-2p)^2}{n}.$$

Compared to part (a), the ratio is

$$\frac{(1-2p)^2}{(1-p)^2}.$$

The baseline-adjusted variance is lower than the unadjusted variance when $p \le 2/3$. It is higher when $p > 2/3$. At $p = 1/2$ the adjusted variance is zero. At $p \to 1$ the adjusted variance is much larger than the unadjusted one. The population-mean baseline is therefore sometimes better and sometimes worse, depending on p.

2.2 Implementing on-policy GRPO

Problem (`tokenize_prompt_and_output`)

Prompt and output tokenization (1 point).

Deliverable: Implement `tokenize_prompt_and_output`.

Implemented in `tests/adapters.py`. Test passes.

Problem (`get_response_log_probs`)

Response log-probs (and entropy) (1 point).

Deliverable: Implement `get_response_log_probs`.

Implemented in `tests/adapters.py`. Test passes.

Problem (`compute_rollout_rewards`)

Computing the rewards of rollouts (1 point).

Deliverable: Implement `compute_rollout_rewards`.

Implemented in `tests/adapters.py`. Test passes.

Problem (`compute_group_normalized_rewards_grpo`)

Group normalization (1 point).

Deliverable: Implement `compute_group_normalized_rewards` with `baseline="mean"`, `advantage_normalizer='`

Implemented in `tests/adapters.py`. Test passes.

Problem (`compute_policy_gradient_loss_on_policy`)

On-policy policy gradient (1 point).

Deliverable: Implement `compute_policy_gradient_loss` with `importance_reweighting_method = "none"`.

Implemented in `tests/adapters.py`. Test passes.

Problem (`aggregate_loss_across_microbatch_sequence`)

Aggregate loss across tokens and sequences (0.5 points).

Deliverable: Implement `aggregate_loss_across_microbatch` with `loss_normalization = "sequence"`.

Implemented in `tests/adapters.py`. Test passes.

Problem (`grpo_train_step_standard_on_policy`)

GRPO train step (5 points).

Deliverable: Implement `grpo_train_step` for the standard on-policy GRPO setting.

Implemented in `tests/adapters.py`. Test passes.

2.3 Experiments

Problem (`grpo_experiments_standard_on_policy`)

Use GRPO to improve OLMo-2-0425-1B performance on GSM8K (10 points).

(a) Training script

Deliverable: A script to run GRPO (standard, on-policy) on GSM8K and OLMo-2-0425-1B.

Pending: script not yet written. Planned location: `scripts/train_grpo.py`. Planned structure:

1. Parse hyperparameters (model name, prompt file, train/val paths, batch sizes, group size, learning rate, max grad norm, sampling parameters).
2. Start the vLLM server on the inference GPU. Set the stop string to `</answer>` for `r1_zero` prompts.
3. Initialize the Hugging Face model on the training GPU. Build the AdamW optimizer.
4. Initialize Weights & Biases. Log hyperparameters.
5. For each of `num_rollout_steps` steps: sync policy weights to vLLM, sample a rollout batch with G completions per prompt, compute rewards, call `grpo_train_step`, log metrics. Every 10 steps run validation on at least 1024 examples and log val rewards. Every 40 steps log a few rollouts for qualitative inspection.
6. Save the model after the final step.

(b) Sanity check ($\sim$50 steps)

Deliverable: Evidence that convinced you the script is correct.

Pending GPU run. The plan is to run the script for $\sim$50 steps and check that validation reward trends upward and that rollouts look sensible.

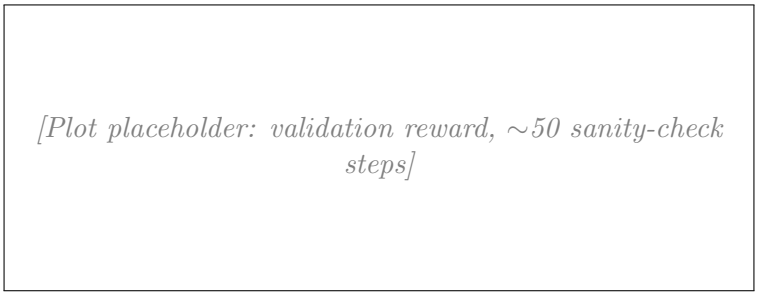
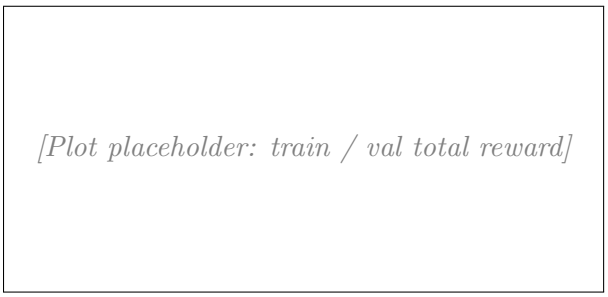


Figure 1: Validation reward, $\sim$50 sanity-check steps. *Pending GPU run.*

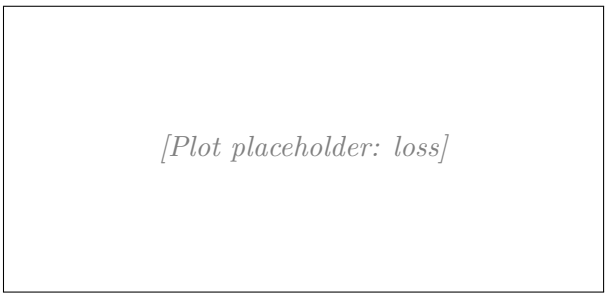
(c) Full run with 4 seeds (r1_zero, suggested hyperparameters)

Deliverable: Plots per metric, qualitative rollout examples, and a procedure that reaches $\geq 25\%$ final validation accuracy.

Pending GPU run. The plan is to run 4 seeds with the suggested hyperparameters and log loss, gradient norm, token entropy, train reward (total and format), val reward (total and format), and val average response length. Plots will show mean and standard deviation across seeds.

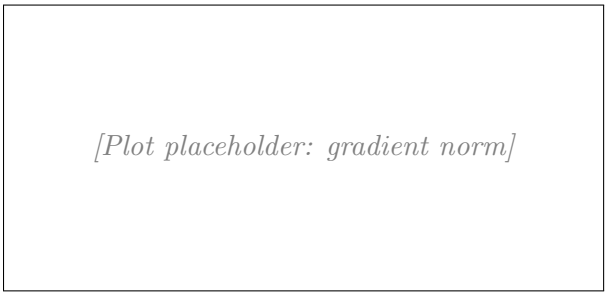


(a) Train / val total reward



(b) Loss

Figure 2: Standard on-policy GRPO, 4 seeds. *Pending GPU run.*



(a) Gradient norm

(b) Token entropy

Figure 3: Diagnostics. *Pending GPU run.*

Rollout examples (before vs. after training). *Pending GPU run.*

Final accuracy. *Pending GPU run.*

Problem (`grpo_learning_rate`)

Tune the learning rate (3 points).

Deliverable: Plot of final validation reward vs. learning rate, with a few sentences of commentary.

Pending GPU run. The plan is to sweep learning rate over $\{3\times10^{-6}, 1\times10^{-5}, 3\times10^{-5}, 1\times10^{-4}\}$ with 2 seeds each. The default is 1×10^{-5}.

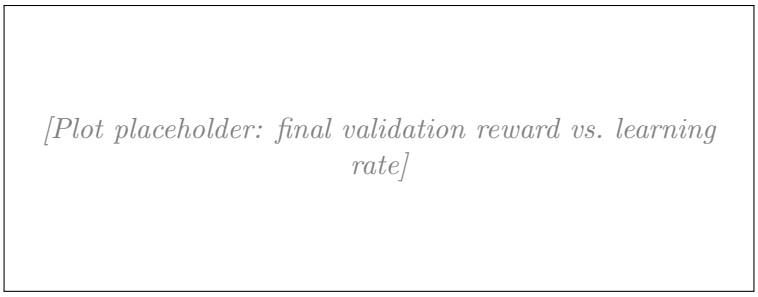


Figure 4: Final validation reward vs. learning rate. *Pending GPU run.*

Problem (`grpo_prompt_ablation`)

Prompt ablation (3 points).

Deliverable: Commentary, supported by plots for referenced metrics.

Pending GPU run. The plan is to run GRPO with three prompts (`question_only`, `r1_zero`, `r1_zero_three_shot`) with the same hyperparameters and a few seeds. Logged metrics will include val reward and token entropy.

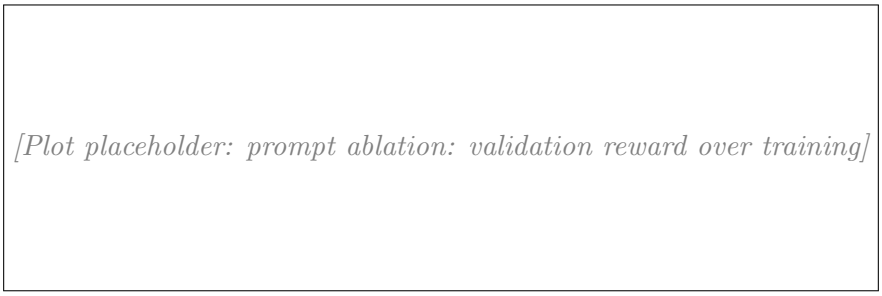


Figure 5: Prompt ablation: validation reward over training. *Pending GPU run.*

3 RL Algorithm Variants

3.1 Dr. GRPO

Problem (`think_about_length_normalization`)

Think about length normalization (1 point).

Deliverable: A few sentences of discussion.

Per-sequence normalization gives every sequence equal weight. A long sequence gets a smaller per-token weight than a short one. This prevents long rollouts from dominating the gradient, but it biases the model toward shorter answers. Constant normalization gives every token equal weight. It is more faithful to the policy gradient, but its scale varies with the total token count in the batch. Per-sequence normalization is better when response lengths vary a lot. Constant normalization is better when you want an unbiased estimate.

Problem (`compute_group_normalized_rewards_drgrpo`)

Dr. GRPO group normalization (0.5 points).

Deliverable: Update `compute_group_normalized_rewards` to support `advantage_normalizer = "none"` and `baseline = "none"`.

Implemented in `tests/adapters.py`. Test passes.

Problem (`aggregate_loss_across_microbatch_constant`)

Dr. GRPO loss aggregation (0.5 points).

Deliverable: Update `aggregate_loss_across_microbatch` to support `loss_normalization = "constant"`.

Implemented in `tests/adapters.py`. Test passes.

3.2 Rejection fine tuning

Problem (`think_about_rft`)

Think about RFT (2 points).

Deliverable: A few sentences of discussion.

Both objectives have the same expectation for binary reward. RFT keeps a $\nabla \log \pi_\theta$ term on correct samples only. Dr. GRPO keeps the same term on all samples but subtracts the group mean. For binary reward $\mathbb{E}_y[\mathbf{1}\{r = 1\}\nabla \log \pi_\theta] = \mathbb{E}_y[r \nabla \log \pi_\theta]$, and the mean subtraction in Dr. GRPO adds zero in expectation. So both estimate $\nabla\eta(x)$. Dr. GRPO has lower variance because the group mean is a valid baseline. RFT skips this correction. I would prefer RFT when most groups are all-correct or all-wrong, since the baseline then does almost nothing and RFT is simpler. I would prefer Dr. GRPO when groups have mixed rewards, since the baseline gives a real variance reduction.

3.3 MaxRL

Problem (`derive_difficulty_reweightings`)

Derive difficulty reweightings induced by different advantage normalizers (6 points).

Setup. Write $\eta(x) = \mathbb{E}_{y\sim\pi_\theta}[r(y|x)]$. The standard policy gradient is $\nabla_\theta J_\theta = \mathbb{E}_x[\nabla\eta(x)]$. The surrogate target is $\nabla_\theta J_{\theta,w} = \mathbb{E}_x[w(x)\nabla\eta(x)]$. The inner expectation over $\sim \pi_\theta(\cdot|x)$ has the identity $\mathbb{E}_y[\nabla \log \pi_\theta(y|x)] = 0$. So for any function $f(x)$ that does not depend on y, $\mathbb{E}_y[f(x)\nabla \log \pi_\theta(y|x)] = 0$.

(a) Dr. GRPO reweighting

Deliverable: An expression in terms of a subset of the problem parameters, with a few sentences of justification.

With $Z = G$ and $G \to \infty$:

$$\frac{1}{G}\sum_{j=1}^{G}(r(y^{(j)}|x) - \mu)\nabla_\theta \log \pi_\theta(y^{(j)}|x) \;\longrightarrow\; \mathbb{E}_y\big[(r - \eta(x))\nabla \log \pi_\theta(y|x)\big].$$

The mean-subtracted term gives zero contribution since $\eta(x)$ does not depend on y. So the limit equals $\mathbb{E}_y[r\nabla \log \pi_\theta(y|x)] = \nabla\eta(x)$. Taking expectation over x:

$$\boxed{w(x) = 1.}$$

Dr. GRPO weights all prompts equally. It matches the unbiased policy gradient in the on-policy limit.

(b) GRPO (with std normalization) reweighting

Deliverable: An expression with derivation.

The estimator divides by the group std. For binary reward,

$$\text{std} \xrightarrow{G\to\infty} \sqrt{\eta(x)(1-\eta(x))}.$$

The numerator converges to $\nabla\eta(x)$ as in (a). Combining:

$$\mathbb{E}_x\Big[\frac{\nabla\eta(x)}{\sqrt{\eta(x)(1-\eta(x))}}\Big] = \mathbb{E}_x[w(x)\nabla\eta(x)] \;\Rightarrow\; \boxed{w(x) = \frac{1}{\sqrt{\eta(x)(1-\eta(x))}}.}$$

The weight is large at $\eta \approx 0$ and $\eta \approx 1$ and small at $\eta = 1/2$. GRPO upweights very easy and very hard prompts.

(c) MaxRL reweighting (divide by group mean)

Deliverable: An expression with derivation.

The estimator divides by μ. With $G \to \infty$, $\mu \to \eta(x)$. So

$$\mathbb{E}_x\Big[\frac{\nabla\eta(x)}{\eta(x)}\Big] = \mathbb{E}_x[w(x)\nabla\eta(x)] \;\Rightarrow\; \boxed{w(x) = \frac{1}{\eta(x)}.}$$

The weight is large for hard prompts (small η) and small for easy prompts. MaxRL upweights hard prompts. Note that the surrogate is $\mathbb{E}_x[\nabla \log \eta(x)]$, i.e. the gradient of the log expected reward.

Problem (`think_about_advantage_normalization`)

Think about advantage normalization (2 points).

Deliverable: A few sentences of discussion.

The three options trade off stability against bias. Std normalization keeps the gradient norm roughly equal across groups, which helps stability, but it upweights very easy and very hard prompts. Mean normalization (MaxRL) upweights hard prompts, which suits curriculum-like training, but it is unstable when the group mean is small. No normalization (Dr. GRPO) is the unbiased policy gradient and weights all prompts equally, but the gradient norm then varies a lot across groups. Use std when training is unstable. Use mean when the dataset has many easy prompts and you want to focus on the hard tail. Use none when you want an unbiased estimate.

Problem (`compute_group_normalized_rewards_maxrl`)

MaxRL group normalization (0.5 points).

Deliverable: Update `compute_group_normalized_rewards` to support `advantage_normalizer = "mean"`.

Implemented in `tests/adapters.py`. Test passes.

3.4 Experiments

Problem (`grpo_train_step_variants_on_policy`)

GRPO train step variants (2.5 points).

Deliverable: Update `grpo_train_step` to support the full range of on-policy variants.

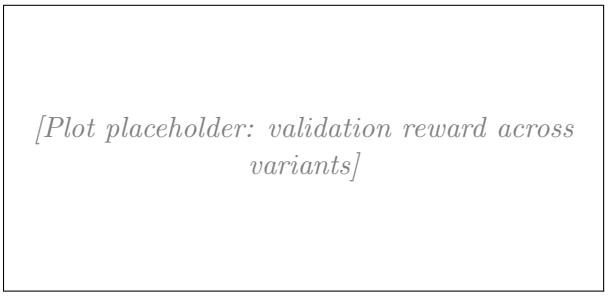
Implemented in `tests/adapters.py`. Tests pass for all four variants. Zero-advantage sequences are skipped before the forward pass to save compute.

Problem (`grpo_experiments_variants_on_policy`)

Compare the performance of different RL algorithms (10 points).

Deliverable: Commentary, supported by plots for referenced metrics.

Pending GPU run. The plan is to run four variants with 4 seeds each and the zero-shot `r1_zero` prompt: `GRPO_constant` (`baseline=mean`, `normalizer=std`, `loss_norm=constant`); `Dr_GRPO` (`baseline=mean`, `normalizer=none`, `loss_norm=constant`); `RFT` (`baseline=none`, `normalizer=none`, `loss_norm=constant`); `MaxRL` (`baseline=mean`, `normalizer=mean`, `loss_norm=constant`).



(a) Validation reward across variants

(b) Token entropy across variants

Figure 6: On-policy variant comparison (4 seeds). *Pending GPU run.*

Discussion. *Pending GPU run.*

4 Off-policy RL

4.1 Importance reweighting / clipping

Problem (`derive_surrogate_objectives`)

Derive surrogate objectives for importance reweighting approaches (2 points).

Deliverable: An expression in terms of a subset of the problem parameters, accompanied by a derivation.

Setup. The pairwise estimator groups tokens into pairs (y_{2t-1}, y_{2t}). For each pair index $t \in \{1, \dots, L/2\}$ define a mixed policy. It samples all pairs from π_{θ_0} except pair t, which is sampled from π_θ:

$$\tilde{\pi}_t(y|x) = \Big(\prod_{s \neq t} \pi_{\theta_0}(y_{2s-1}|x, y_{<2s-1})\pi_{\theta_0}(y_{2s}|x, y_{<2s})\Big) \cdot \pi_\theta(y_{2t-1}|x, y_{<2t-1})\pi_\theta(y_{2t}|x, y_{<2t}).$$

Surrogate. The surrogate objective sums the reward under each mixed policy:

$$\boxed{J_\theta^{\text{pair}} = \mathbb{E}_x\left[\sum_{t=1}^{L/2} \mathbb{E}_{y\sim\tilde{\pi}_t}\big[r(y|x)\big]\right].}$$

Derivation. Take the gradient and reweight each term so it is sampled from π_{θ_0} everywhere. For a fixed pair t, the ratio of $\tilde{\pi}_t$ to π_{θ_0} is exactly the pair t ratio:

$$\frac{\tilde{\pi}_t(y|x)}{\pi_{\theta_0}(y|x)} = \frac{\pi_\theta(y_{2t-1}|x, y_{<2t-1})\pi_\theta(y_{2t}|x, y_{<2t})}{\pi_{\theta_0}(y_{2t-1}|x, y_{<2t-1})\pi_{\theta_0}(y_{2t}|x, y_{<2t})}.$$

Applying the log-derivative trick on the pair t factors only (the other factors do not depend on θ):

$$\nabla_\theta \mathbb{E}_{y\sim\tilde{\pi}_t}[r(y|x)] = \mathbb{E}_{y\sim\pi_{\theta_0}}\Big[\tfrac{\tilde{\pi}_t(y|x)}{\pi_{\theta_0}(y|x)}\, r(y|x)\, \nabla_\theta \log\big(\pi_\theta(y_{2t-1}|x, y_{<2t-1})\pi_\theta(y_{2t}|x, y_{<2t})\big)\Big].$$

Summing over t and over x recovers the given pairwise estimator. So the surrogate above is the objective whose gradient equals that estimator.

4.2 Implementation

Problem (`compute_policy_gradient_loss_off_policy`)

Off-policy policy gradient with token-level reweighting (1 point).

Deliverable: Update `compute_policy_gradient_loss` to support `importance_reweighting_method = "noclip"` or `"grpo"`.

Implemented in `tests/adapters.py`. Tests pass for both `noclip` and `grpo`.

Problem (`compute_policy_gradient_loss_off_policy_gspo`)

Off-policy policy gradient with sequence-level reweighting (1 point).

Deliverable: Update `compute_policy_gradient_loss` to support `importance_reweighting_method = "gspo"`.

Implemented in `tests/adapters.py`. Test passes. The geometric mean is computed in log space to stay numerically stable.

Problem (`grpo_train_step_off_policy`)

Off-policy GRPO train step (2.5 points).

Deliverable: Update `grpo_train_step` to support off-policy arguments.

Implemented in `tests/adapters.py`. Tests pass for `noclip`, `grpo`, and `gspo`.

4.3 Experiments

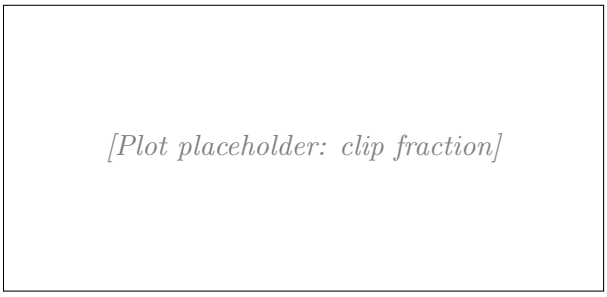
Problem (`grpo_experiments_off_policy`)

Compare the performance of different off-policy algorithms (10 points).

Deliverable: Commentary, supported by plots for referenced metrics.

Pending GPU run. The plan is to run four off-policy variants with 4 seeds each and the zero-shot `r1_zero` prompt, with `rollout_batch_size=256`, `train_batch_size=8`, `gradient_accumulation_steps=1`: `offpolicy_naive` (`importance_reweighting_method=none`); `offpolicy_noclip` (`noclip`); `offpolicy_clip` (`grpo`, $\varepsilon = 0.2$); `offpolicy_gspo` (`gspo`, $\varepsilon = 3\times10^{-4}$). Logged metrics include clip fraction.

(a) Validation reward across off-policy variants



(b) Clip fraction

Figure 7: Off-policy variant comparison (4 seeds). *Pending GPU run.*

Discussion. *Pending GPU run.*

Problem (`think_about_importance_reweighting`)

Think about importance reweighting (2 points).

Deliverable: A few sentences of commentary.

No reweighting has the highest bias and the lowest variance. It works when π_θ and π_{θ_0} are close, but it breaks down once they drift apart. PPO/GRPO token-level clipping sits in the middle. It caps each ratio in $[1-\varepsilon, 1+\varepsilon]$ to control variance, at the cost of bias on clipped tokens. This is the standard choice for off-policy LLM RL. GSPO clips a sequence-level geometric-mean ratio. It is better when sequences are long and credit belongs at the sequence level. Pick no reweighting for nearly on-policy training, token-level clipping for the standard off-policy case, and GSPO for long sequences.

5 Try your own policy gradient estimator

Problem (`try_your_own`)

Try your own policy gradient estimator (10 points).

Deliverable: Commentary, plots, and optionally mathematical derivations.

Proposed estimator: leave-one-out baseline (LOO-GRPO)

The standard GRPO baseline is the group mean. The group mean includes the very sample it is used to adjust. The leave-one-out (LOO) baseline drops that sample. For prompt i and rollout j define

$$\mu_i^{(j)} = \frac{1}{G-1} \sum_{k \neq j} r(y^{(i,k)}|x^{(i)}).$$

The estimator is otherwise identical to Dr. GRPO (constant normalization, no std):

$$\hat{g}_{\text{LOO}} = \frac{1}{Z} \sum_{i=1}^{B} \sum_{j=1}^{G} \sum_{t=1}^{\text{len}(y^{(i,j)})} \left(r(y^{(i,j)}|x^{(i)}) - \mu_i^{(j)} \right) \nabla_\theta \log \pi_\theta\big(y_t^{(i,j)} \mid x^{(i)}, y_{<t}^{(i,j)}\big).$$

This changes one thing relative to Dr. GRPO: the baseline.

Theoretical motivation

The group mean is correlated with the sample it adjusts. The LOO baseline is independent of sample j by construction. So the LOO estimator is unbiased even at finite G, while the Dr. GRPO estimator carries an $O(1/G)$ bias.

For binary reward, the difference is concrete. The Dr. GRPO advantage for a correct rollout is $1-\mu_i$ where μ_i already counts that rollout's reward. The LOO advantage is $1-\mu_i^{(j)} = 1-(\mu_i G - 1)/(G-1)$, which is strictly larger. So LOO gives a larger gradient signal on rare correct rollouts and a stronger penalty on rare incorrect ones. This matches the intuition that the policy gradient should care most about samples that deviate from the rest of the group.

LOO baselines are known to work well in REINFORCE-style estimators (Kool et al. 2019; Richter et al. 2020). The cost is a tiny extra computation per group (still $O(G)$).

Results

Awaiting training runs. The plan is to run LOO-GRPO against Dr. GRPO at 4 seeds with identical hyperparameters on OLMo-2-0425-1B and GSM8K. Logged metrics will include loss, gradient norm, token entropy, train reward, and validation reward.

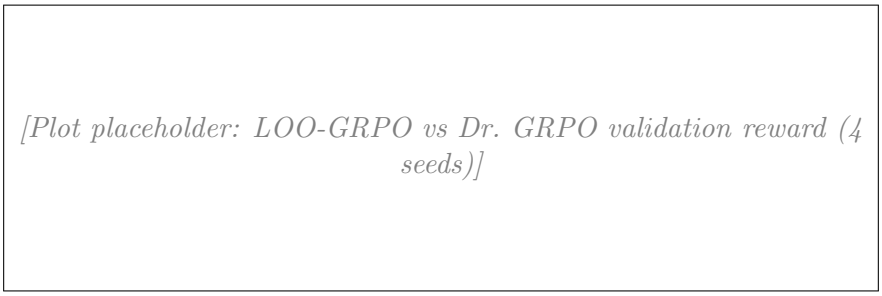


Figure 8: Custom estimator vs. baselines (validation reward). *Pending GPU run.*

Does it beat the baselines? *Pending experimental results.*